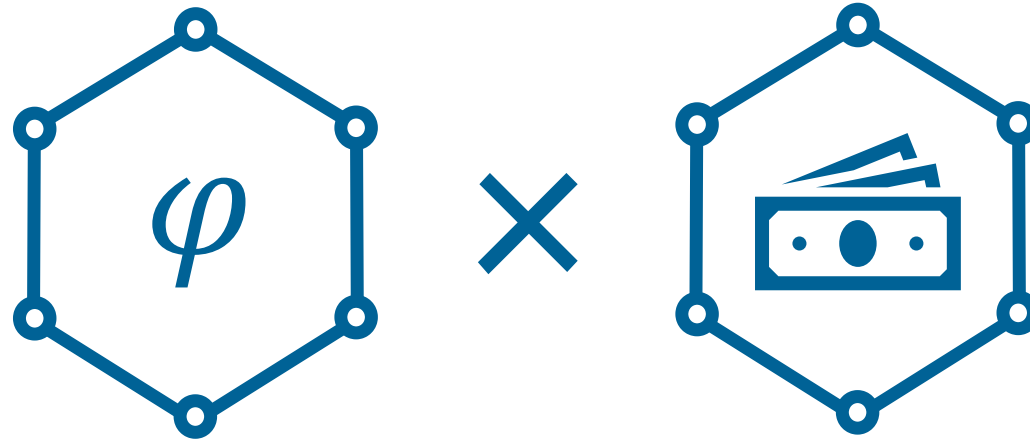
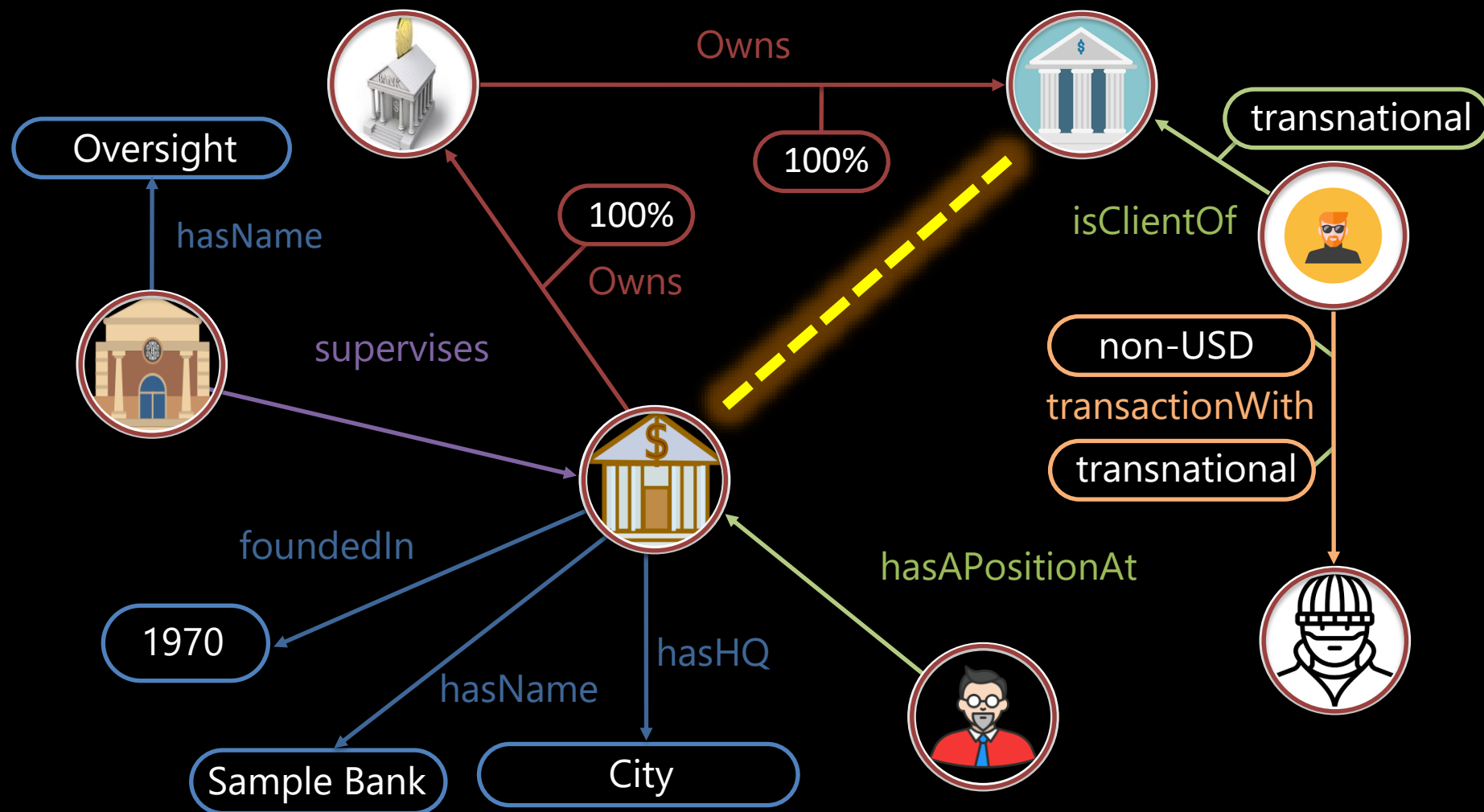


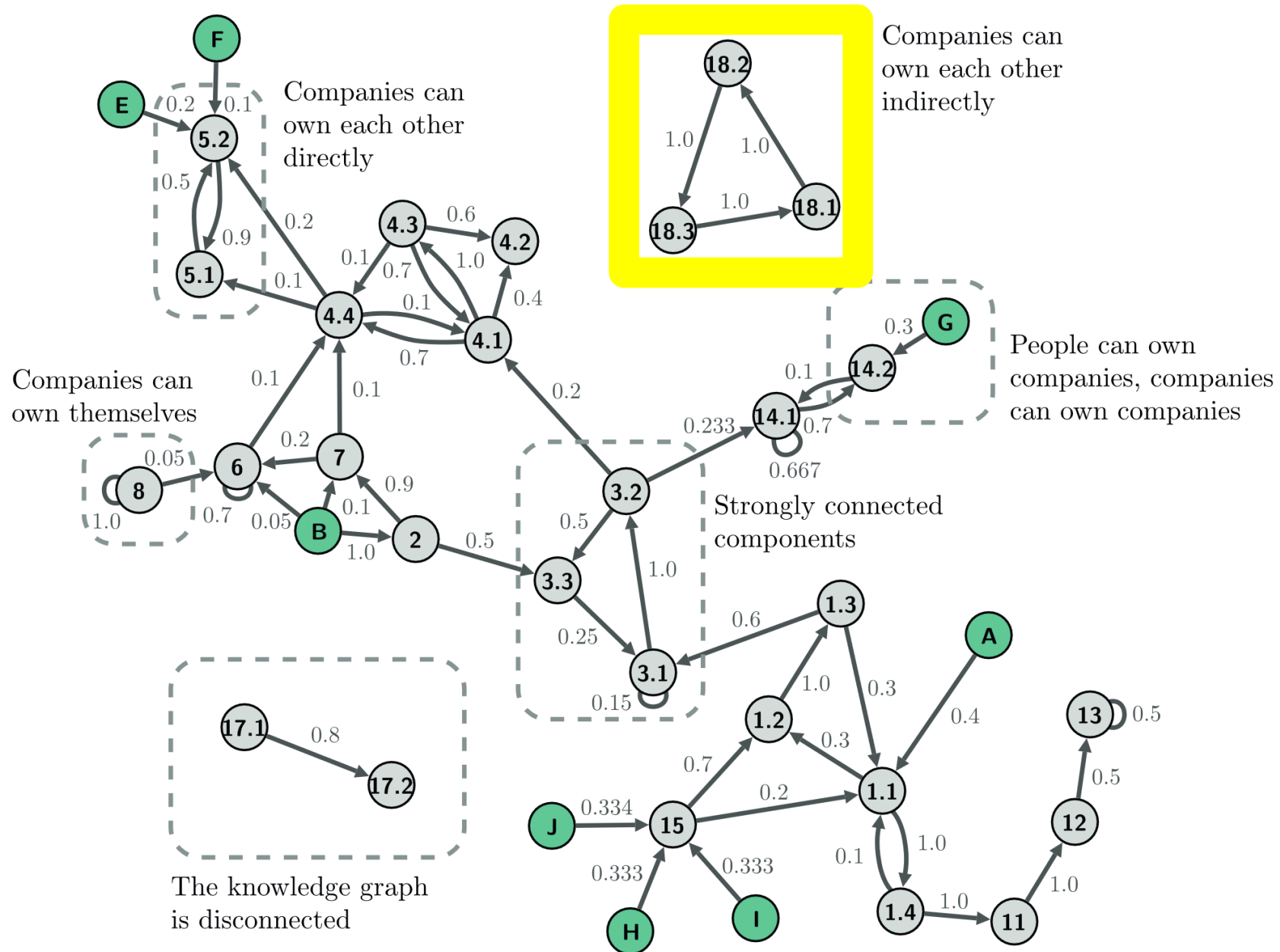


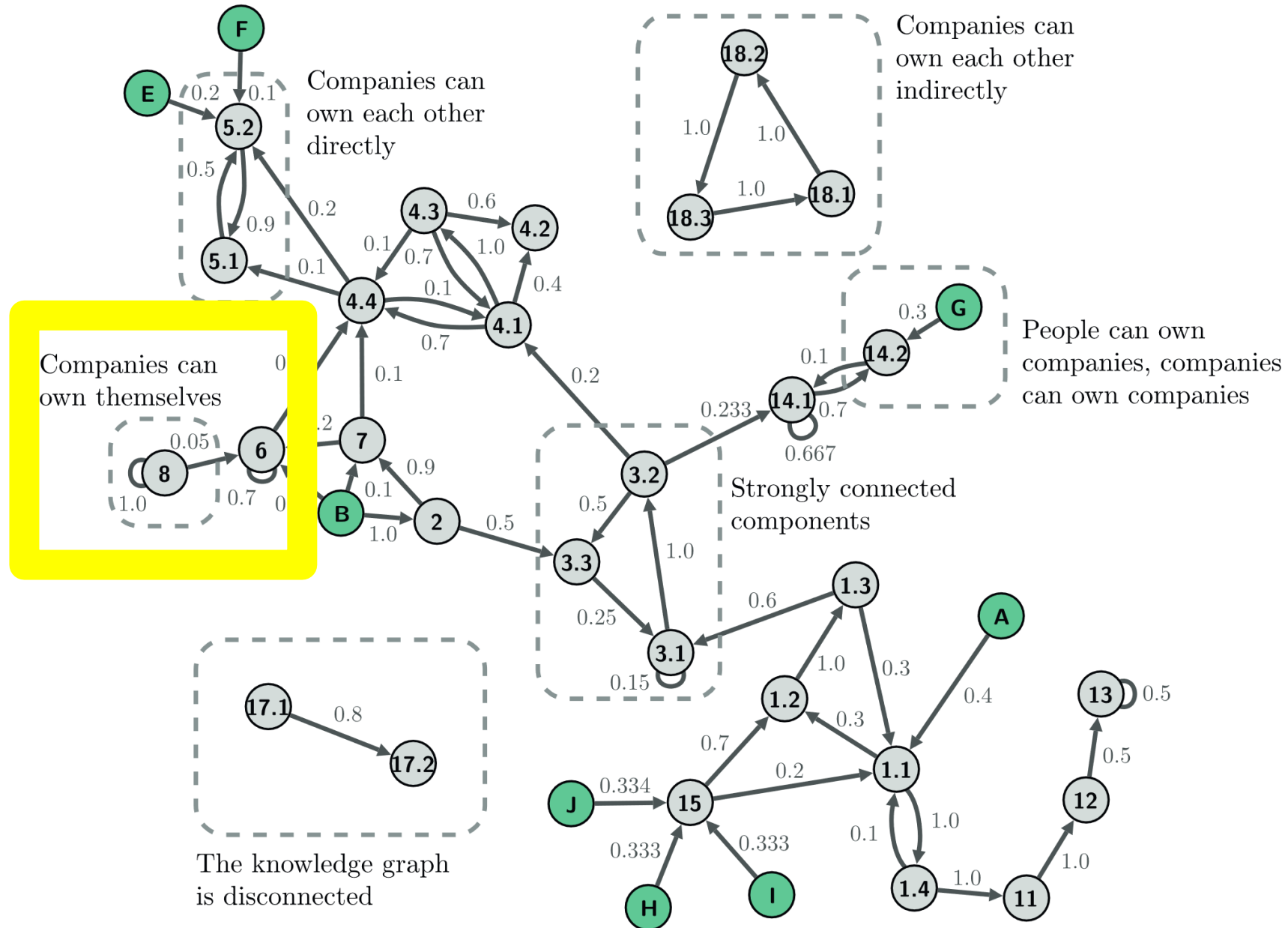
Logical Knowledge in KGs

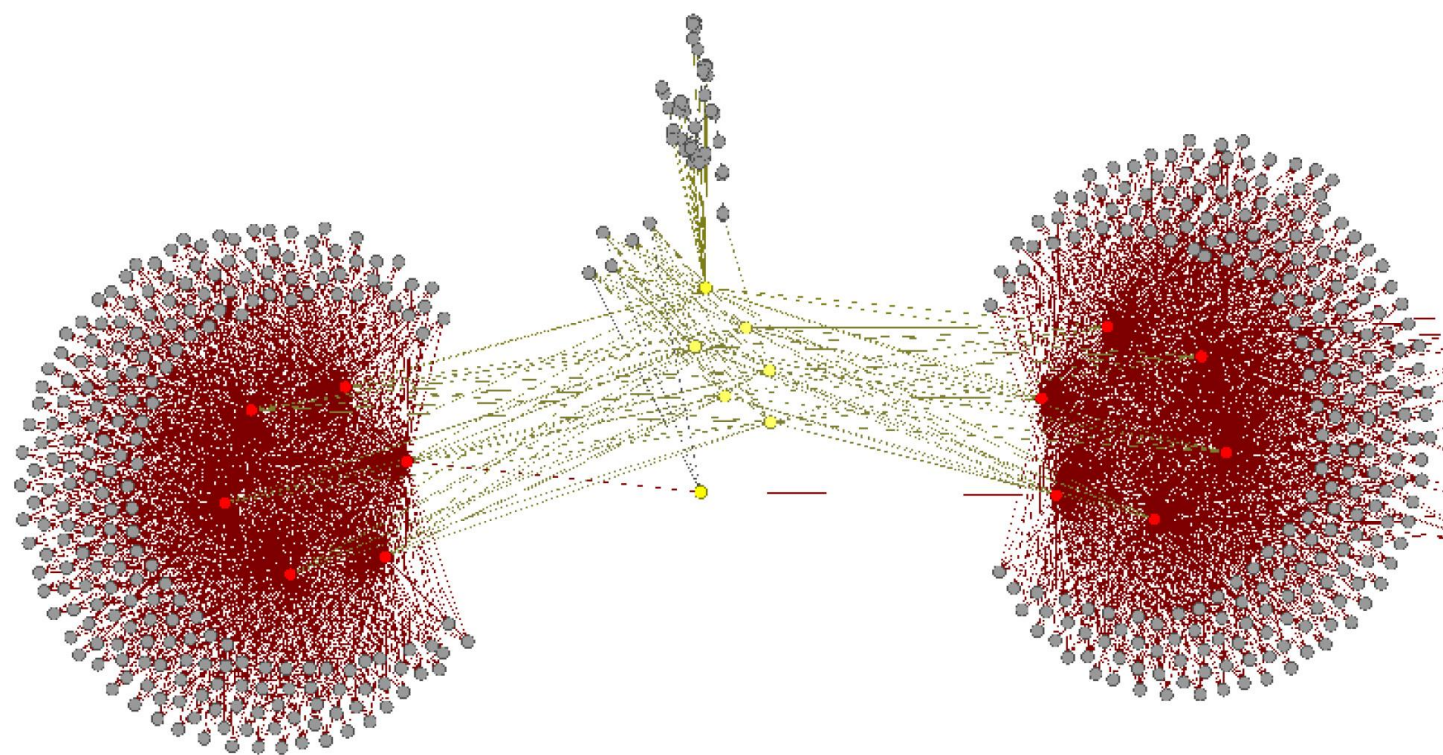
Recursion in the Real World

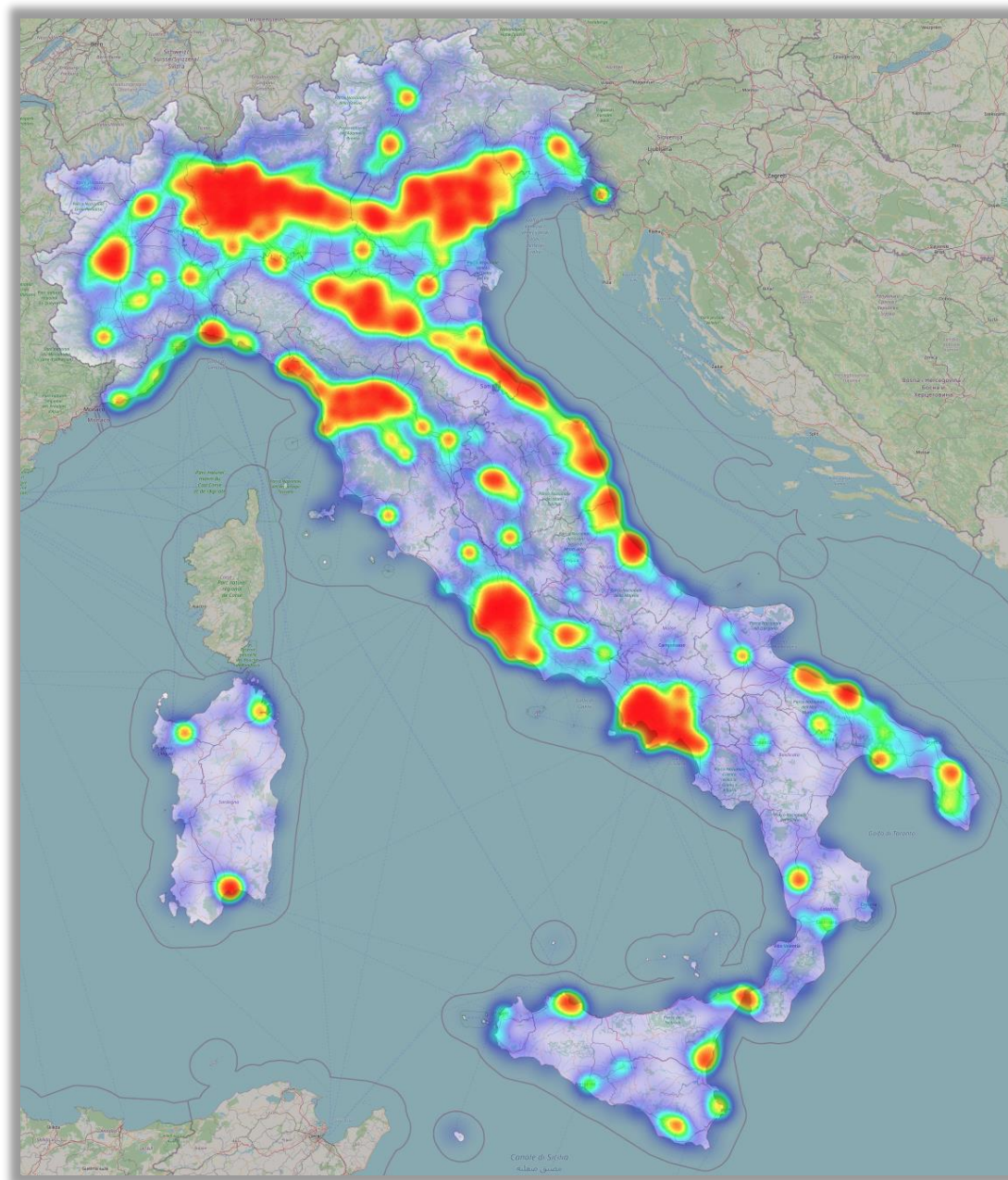
Emanuel Sallinger









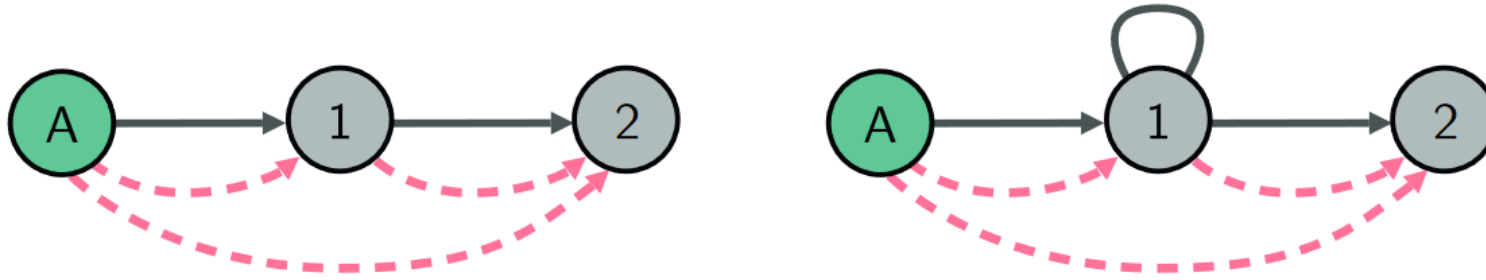




Ownership and Control

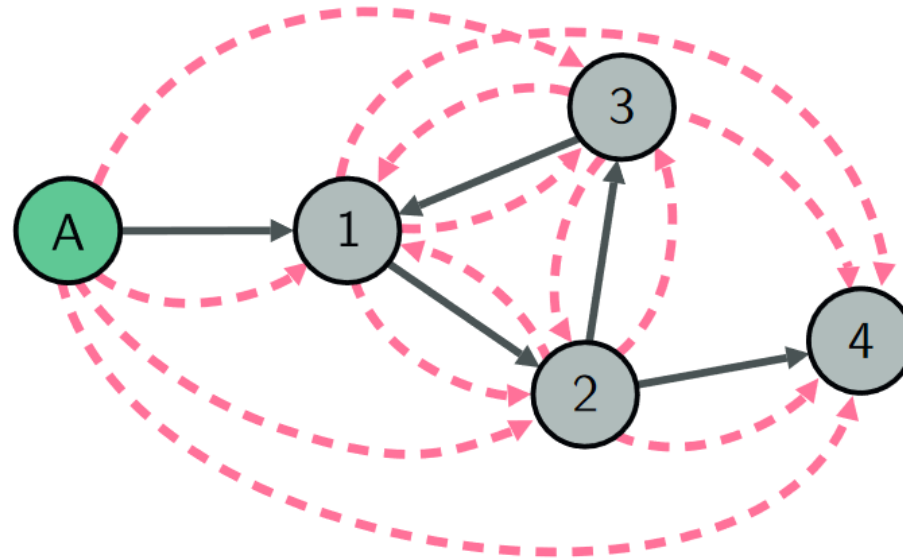


Ownership



(a) A simple indirect ownership.

(b) Indirect ownership with a self-loop.



(c) Indirect ownership with a strongly connected component.

Ownership

Definition 1. An ϵ -Baldone path P from s to t is a path $[s, p_1, \dots, p_n, t]$ such that $s \neq p_i$ for $i = 1, \dots, n$ and $w(P) > \epsilon$, with $\epsilon \in \mathbb{R}^+$ and $0 < \epsilon \leq 1$. We denote the weight of an ϵ -Baldone path as $w_\epsilon(P)$.

Definition 2. The ϵ -Baldone ownership of a company s on a company t in a graph G is a function $O_\epsilon^G(s, t) : (N \times N) \rightarrow \mathbb{R}^+ \cup \{\infty\}$ defined as $(s, t) \rightarrow \sum_{P_i \in B_\epsilon} w_\epsilon(P_i)$, where B_ϵ is the set of all possible ϵ -Baldone paths from s to t .

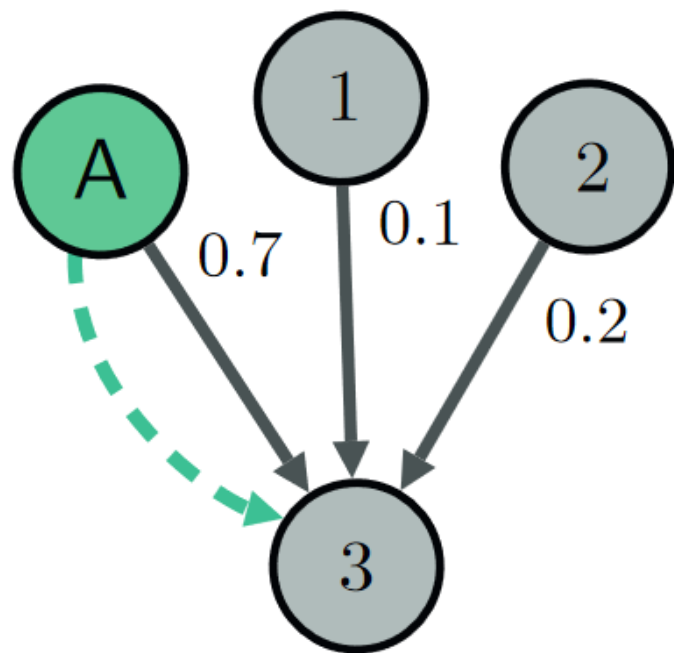
Definition 3. The Baldone ownership (which we will also refer to as integrated ownership) of a company s on a company t in a graph G is a function $O^G(s, t) : (N \times N) \rightarrow \mathbb{R} \cup \{\infty\}$ defined as $(s, t) \rightarrow \lim_{\epsilon \rightarrow 0} O_\epsilon(s, t)$.



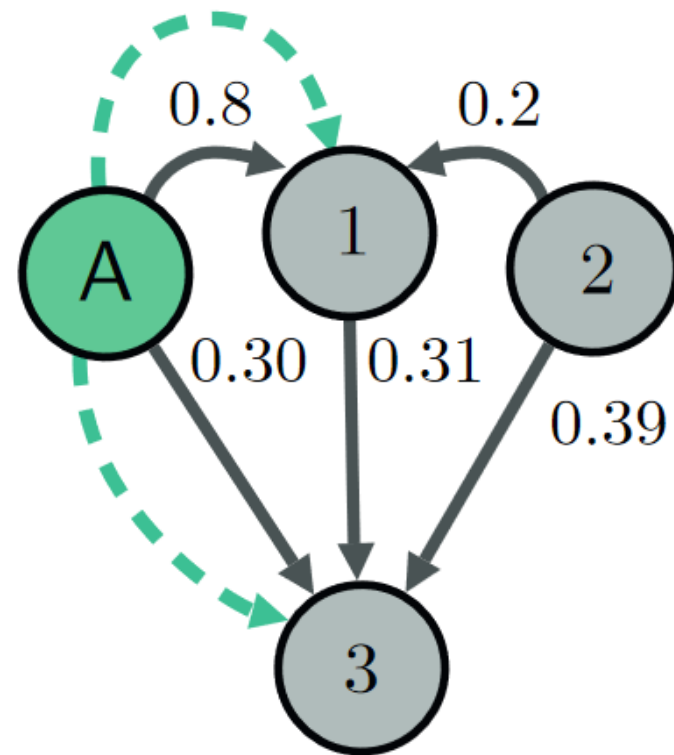
$$Own(x, y, w), w > \epsilon, v = sum(w), p = [x, y] \rightarrow IOwn(x, y, v, p). \quad (1)$$

$$IOwn(x, z, w_1, p_1), IOwn(z, y, w_2, p_2), p = p_1 | p_2, BaldonePath(p, v, \epsilon), \\ v = sum(w_1 \times w_2), \rightarrow IOwn(x, y, v, p). \quad (2)$$

Control



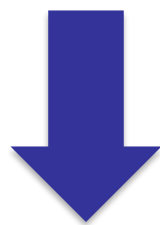
(a) Direct control.



(b) Indirect control.

Control

Definition 4. *A company (or a person) x controls a company y , if: (i) x directly owns more than 50% of y ; or, (ii) x controls a set of companies that jointly (i.e., summing the shares), and possibly together with x , own more than 50% of y .*



$$Control(x) \rightarrow Control(x, x) \quad (1)$$

$$Control(x, y), Own(y, z, w), v = msum(w, \langle y \rangle), v > 0.5 \rightarrow Control(x, z) \quad (2)$$



Golden Power



Golden Power Check

Golden Power Check

- Goal:* The general goal is checking whether an acquisition (of shares, stocks, etc.) causes any strategic Italian company to become controlled by a foreign company.
- Setting:* Let S be a set of strategic companies and F be a set of foreign companies. Let t be a transaction (e.g., an offer issued by a company x to buy an amount s of shares of a company y), with $x, y \in S \cup F$.
- Question:* Decide whether t causes any company in F to control a company in S .
- Insight:* Consider exerting “golden power” to block t .

Golden Power Check

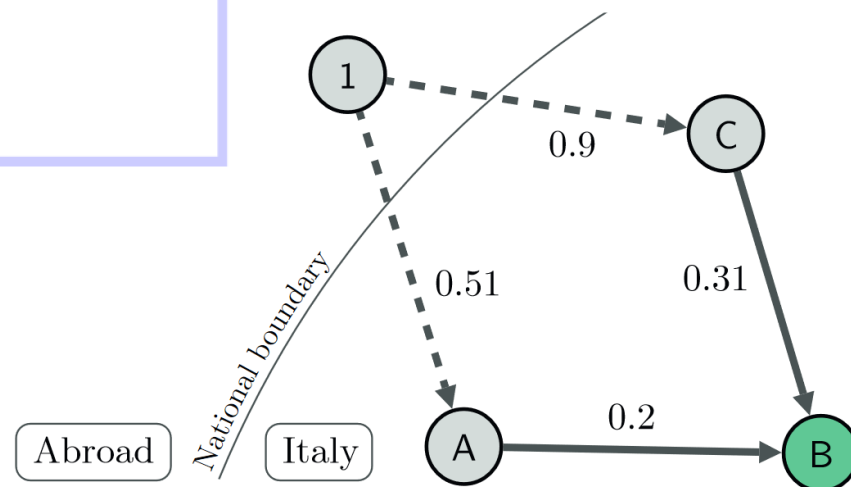
Golden Power Check

Goal: The general goal is checking whether an acquisition (of shares, stocks, etc.) causes any strategic Italian company to become controlled by a foreign company.

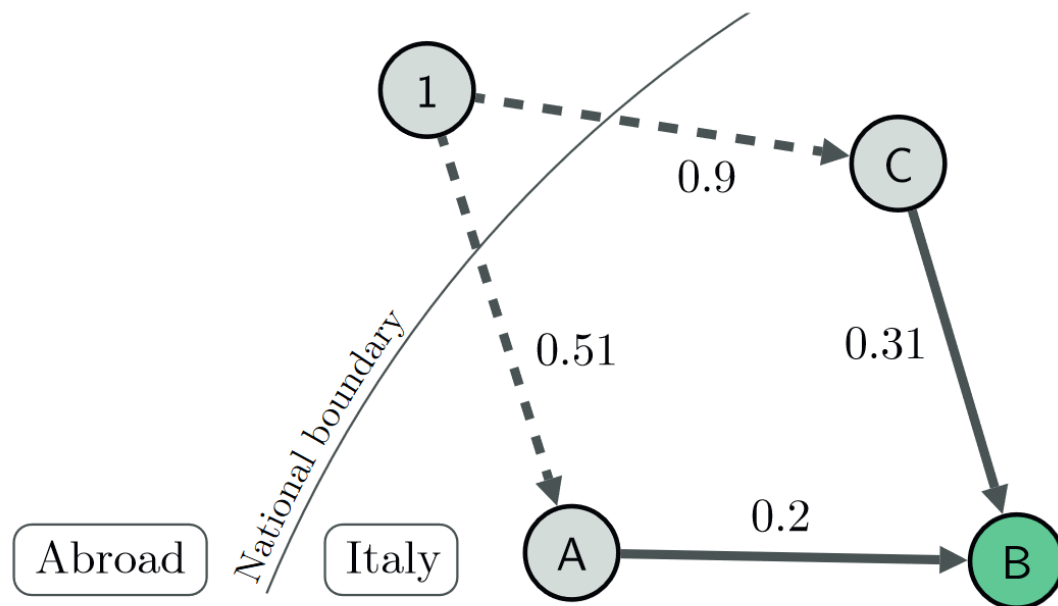
Setting: Let S be a set of strategic companies and F be a set of foreign companies. Let t be a transaction (e.g., an offer issued by a company x to buy an amount s of shares of a company y), with $x, y \in S \cup F$.

Question: Decide whether t causes any company in F to control a company in S .

Insight: Consider exerting “golden power” to block t .



Golden Power Check

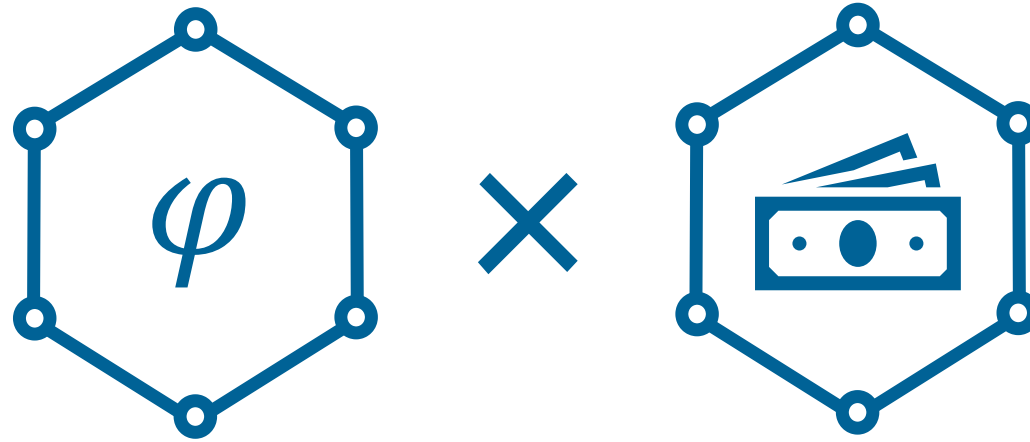


$$T(x, y, w) \rightarrow Own(x, y, w) \quad (1)$$

$$Control(x, y) \rightarrow Control(x, x) \quad (2)$$

$$Control(x, y), Own(y, z, w), v = msum(w, \langle y \rangle), v > 0.5 \rightarrow Control(x, z) \quad (3)$$

$$F(x), S(y), Control(x, y) \rightarrow GPCheck(x, y) \quad (4)$$



Logical Knowledge in KGs

Recursion in the Real World

Emanuel Sallinger